Swajan Barua

Junior Software Engineer

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PROFESSIONAL SUMMARY

Recent Computer Science and Engineering graduate with a strong foundation in backend development and scalable system architecture. Demonstrated expertise in building end-to-end web and mobile applications using Go, NestJS, NextJS, and Kotlin. Experienced in designing and managing diverse database architectures, including PostgreSQL, MySQL, and Firebase. Passionate about solving real-life problems through creative solutions, with hands-on experience in designing secure RESTful APIs, managing diverse database architectures, and automating complex workflows. Interested in contributing emerging technology stacks to build smarter and more efficient systems while contributing to open-source development and technical innovation. Committed to continuous learning and staying updated with the latest software development trends, architectural patterns, and modern frameworks to deliver high-quality code.

EDUCATION

Bachelor of Science in Computer Science & Engineering — **2022 – 2026**
American International University-Bangladesh (AIUB) — *Dhaka, Bangladesh*

- **Major: Software Engineering**
- **Relevant Coursework:** Advanced Programming in Web technologies, Mobile Application Development, Database Systems, Advanced Programming with .NET, Machine Learning, Software Engineering, Data Structures & Algorithms.
- **Thesis:** A Comparative Performance Analysis of Machine Learning Models for Thyroid Disease Classification.
- **Academic Focus:** Problem Solving, Full-Stack Web Development, Mobile Application Development, and Efficient Data Structure Implementation.

Higher Secondary Certificate (Science) — **2018 – 2020**
Milestone College — *Dhaka, Bangladesh*

Secondary School Certificate (Science) — **2013 – 2018**
Harishchar Union High School & College — *Cumilla, Bangladesh*

TECHNICAL SKILLS

- **Programming Languages:** Go, C#, Kotlin, C, C++, Java, Python, JavaScript, PHP.
- **Frameworks & Libraries:** Node.js, React.js , NestJS, Next.js, Gin, .NET Core.
- **Databases:** PostgreSQL, MySQL, Firebase.
- **Tools & Technologies:** Git, GitHub, Postman, Chrome DevTools, Android Studio.
- **System Integration & Security:** JWT Authentication, Role-Based Access Control (RBAC), OTP Verification Systems, SMTP.
- **Competitive Programming:** Codeforces (Max Rating: 1162), LeetCode.

PROJECTS & PRACTICAL EXPERIENCE

ArkPlayZone — Nov 2025
Academic Project | *Next.js, NestJS, PostgreSQL, JWT, SMTP, OTP, RESTful API*

- Developed a full-stack booking system with real-time slot tracking, pre-booking, and automated cancellation logic using **NestJS** and **PostgreSQL**.
- Implemented a secure identity layer featuring **JWT**, **RBAC**, and a custom **OTP/SMTP** verification workflow for user registration and notifications.
- Optimized backend reliability to handle concurrent booking requests and ensure data integrity.

Bkash Personal Gateway & Transaction Verifier Dec 2025
Personal Learning Project | *Kotlin, NestJS, Next.js, PostgreSQL, JWT*

- Engineered a custom verification ecosystem to automate transaction tracking for personal accounts, bypassing the need for official merchant APIs.
- Built an **Android background service** to intercept/parse SMS notifications and sync data with a **NestJS** backend for real-time payment matching.
- Developed a **Next.js** dashboard with JWT-protected endpoints to securely visualize and manage transaction history.

AiubBuddy Jan 2026
Academic Project | *Kotlin, NestJS, PostgreSQL, Firebase (FCM/Auth), SQLlite, Retrofit*

- Developed an event-driven **Android app** using **MVVM** to manage academic routines and faculty data for 1,000+ potential users.
- Built a **NestJS** notification engine using **FCM** to push real-time university notices even when the application is closed.
- Implemented an **offline-first** architecture using **Room (SQLite)** for local data persistence and Firebase for secure OTP authentication.

Codeforces Activity Tracker – Automation Bot Dec 2024
Personal Learning Project | *Go (Golang), Codeforces API, Telegram Bot API*

- Developed a CLI-based monitoring tool in **Go** to track competitive programming progress and automate daily performance reporting.
- Built a real-time data pipeline that fetches and filters submission data from the **Codeforces API**, analyzing user solve rates and difficulty metrics.
- Connected the **Telegram Bot API** to deliver formatted Markdown reports and instant notifications to a dedicated chat at scheduled intervals.

ACHIEVEMENTS

- **Competitive Programming:** Ranked **17th out of 100+** participants in the AIUB CS Fest Programming Contest (2024).
- **Academic Excellence:** Awarded **Dean's List Honors** for the Fall 2023–24 semester in recognition of outstanding academic performance.
- **Robotics:** Advanced to the **2nd Round** of the AIUB CS Fest Robo Soccer Competition (2024), demonstrating effective team collaboration and logical strategy.

TOOLS & TECHNICAL PROFICIENCIES

- **Data Formats:** JSON, CSV, Markdown
- **Version Control:** Git & GitHub
- **Scripting:** Bash, PowerShell, Regex
- **Environment:** Linux/Unix CLI